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Technical Information Document

DXT Data Management System Data Interface for AMICUS

The purpose of this document is to define the interface specification required for communication between DXT and external Data Management System (DMS).

This specification defines the physical interface, data communication protocol, and data content of messages passed between DXT and the DMS.

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Chapter 1 General Description

DXT is built using REST principles which ensure predictable URLs that make writing interface easy. The DXT interfaces follow HTTP rules, enabling a wide range of HTTP clients and servers to be used to interact with DXT.

This chapter gives an overview of DXT interface structure.

- Overview
- Web Services
  - Procedure Parameters
  - Procedure Parameters Status
  - Electronic Record
- Message Format
  - HTTP Request
  - HTTP Response
  - Root Element

Section 1.1 Overview

DXT implements bi–directional communications with external data management systems (DMS) to provide asynchronous communications. Based on the type of the message, the communications can be initiated by either DXT or the DMS.

For example, when the DMS sends procedure setup file to DXT, the DMS will act as a client posting setup data to the DXT server. Likewise, when DXT sends an electronic record to the DMS, DXT will act as a client posting procedure data to the DMS server.

Section 1.2 Web Services

This section provides an overview of the web services included with DXT. The following chapters contain detailed description of each service as well as structure and content of each message.

Procedure Parameters

The DMS will use the Procedure Parameters service to send a procedure setup file to DXT.

The procedures setup parameters include details about the donor to enable the instrument to determine the appropriate procedure settings for the donor.

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The main purpose of remote procedure setup is to help improve procedure setup time and help reduce manual entry errors at the instrument. DXT uses push method to achieve this.

Push Method

In the push model of remote procedure setup, the DMS will send the instrument status request to determine availability of the instruments for the remote procedure setup. The DMS will select an instrument and send the procedure setup file to DXT, then DXT will send the procedure setup file to the instrument specified by the DMS. This is commonly referred to as the push model since DXT is pushing the data down to the instrument.

The following sequence diagram depicts interactions between DMS, DXT, instrument, and operator for the push model.

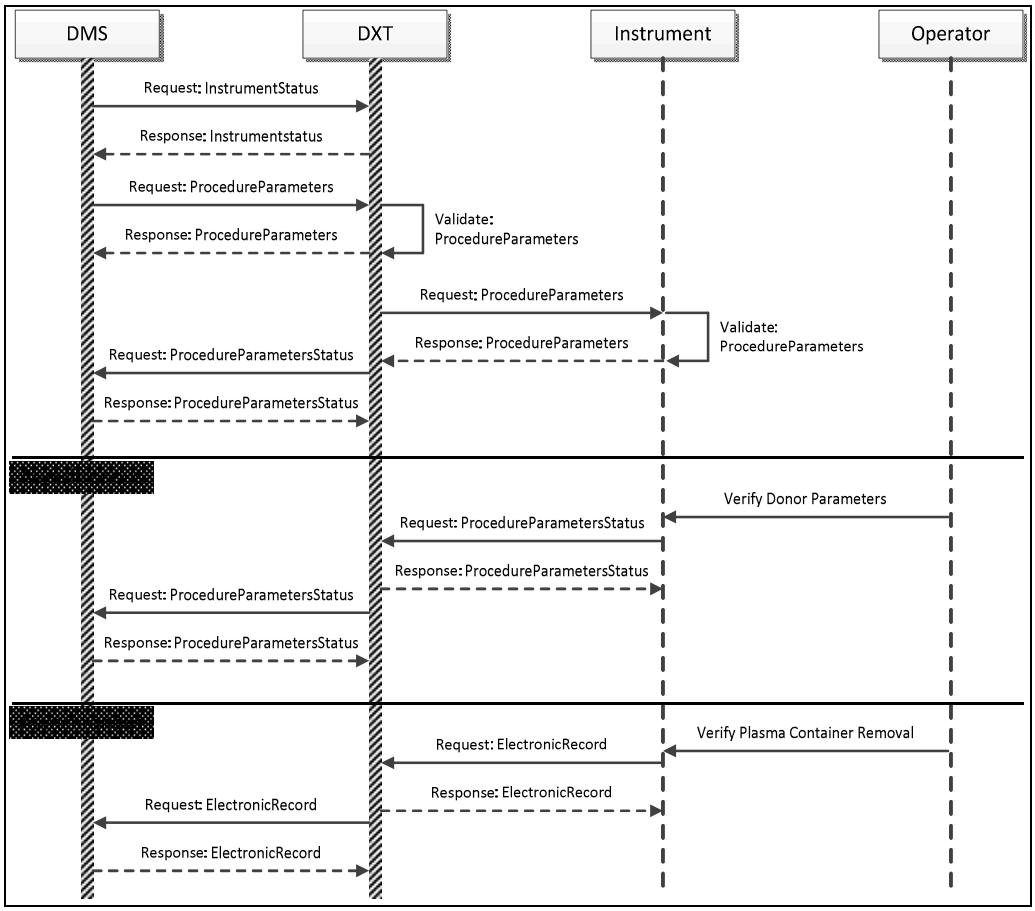


Figure 1 Typical Sequence of Interactions in PUSH Mode

Procedure Parameter Status

DXT will use the Procedure Parameter Status service to notify the DMS of the status of the procedure parameter.

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Electronic Record

DXT will use the Electronic Record service to provide the procedure record data to the DMS.

DXT receives and stores data from connected instruments. Using the Electronic Record service, DXT can forward the data to the DMS. The data maintained in DXT can be considered as the 21 CFR Part 11 and/or EU Annex 11 compliant electronic records for a blood or platelet center, provided the center has policies and procedures for system usage as described in the regulation. Refer to DXT Data Management System Administrator's Guide Chapter 7 – Cyber Security, Part 11, and Annex 11 for considerations for deploying DXT in a 21 CFR Part 11 and/or EU Annex 11 compliant environment.

Instrument Status

The DMS will use the Instrument Status service to determine the availability of the site's instruments for remote procedure setup.

Section 1.3 Message Format

HTTP Request

When DXT acts as a client, it will send all HTTP request headers in the following format. When DXT acts as a server, it expects all HTTP request headers in the same format.

```
METHOD URL HTTP/1.1 Host: SERVER_ADDRESS
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Connection: Keep-Alive
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

Method must be either GET or POST, DXT does not currently support resources updates (PUT) or deletion (DELETE). DXT will return an error response to HTTP PUT and HTTP DELETE requests.

HTTP header **Host** must be present and contain network name of the destination server.

HTTP Header **Content–Length** must be present and contain payload (XML) length in bytes.

HTTP Header **Content–Type** must be "text/xml; charset=utf–8", unless otherwise noted for the specific message.

HTTP Header **Date** must be present and contain date and time when message was generated using full Date format, specified by RFC–2616, Hypertext Transfer Protocol – HTTP/1.1. For example:

```
Date: Fri, 14 Jun 2013 21:12:59 GMT
```

DXT expects all messages to be received within 2 minutes to ensure proper functionality. DXT will respond with an error to any message that contains an HTTP header Date Time more than 2 minutes in the past or in the future from DXT server time.

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HTTP Header must include **Connection: Keep–Alive** directive.

HTTP Header must include **Cache–Control** with cache directive set to “no–cache” and “no–transform”. Comma “, ” and additional values (e.g., “private”) are acceptable.

The value of HTTP Header **FK–SourceId** and **FK–DestinationId** must be set as following:

For DXT, it should be “0”.
For DMS, it should be “1”.
For QC System, it should be “2”.

DXT expects **FK–SourceId** in HTTP Header of all inbound messages, and will supply **FK–DestinationId** in HTTP Header of all outbound messages.

For example, when the DMS sends data to DXT as a server, the IDs will be as following:

```
FK-SourceId: 1
FK-DestinationId: 0
```

Section 1.4 Root Element

All XML messages must have a root element of DMSMessage as shown below. No data is associated with this element, it is a container for all child elements.

```
<xsd:element name="DMSMessage">
    <xsd:complexType>
        <xsd:sequence>

            <CHILD ELEMENTS>

            </xsd:sequence>
</xsd:complexType>
</xsd:element>
```

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Chapter 2 DXT Server

This chapter defines messages that DXT accepts from external data management systems.

For possible error responses from DXT refer to Error Handling section at the end of this chapter.

Section 2.1 Procedure Parameters

The DMS will use the Procedure Parameters service to send procedure parameters to DXT. This enables the remote procedure setup if DXT is licensed for this feature.

Procedure Parameters URL

```
POST /fresenius-kabi/dxt/dms/procedure-parameters HTTP/1.1
```

Procedure Parameters request to DXT must supply ProcedureParameters XML object.

Procedure Parameters Content Type

Procedure Parameters request to DXT must contain the following content type header element:

```
Content-Type: multipart/mixed;
boundary=this_is_fresenius_kabi_http_boundary
```

The first part of the message must have content type “text/xml; charset=utf–8” and XML parameters file. Other parts are reserved for future use and should not be supplied:

```
--this_is_fresenius_kabi_http_boundary
Content-Type: text/xml; charset=utf-8

<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <ProcedureParameters>

        ...

    </ProcedureParameters>

</DMSMessage>
--this_is_fresenius_kabi_http_boundary--
```

ProcedureParameters XSD Definition

This complex type defines the contents of the Procedure Parameters structure as shown below.

```
<xsd:complexType name="ProcedureParameters">
    <xsd:sequence>

        <xsd:element name="DonationSetupId" minOccurs="1" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
```


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```
                    <xsd:pattern value="[a-zA-Z0-9]*" />
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="InstrumentSerialNumber" minOccurs="0">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="20" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="SiteIdentifier" minOccurs="1" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="20" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="DXTProcedureParameters" type="xsd:anyType" />

    </xsd:sequence>
</xsd:complexType>
```

ProcedureParameters Data Description

The following table explains the content of Procedure Parameters message:

Element	Parent Element	Data Type	Description
ProcedureParameters	DMSMessage		Container. This tag is required.
DonationSetupId	ProcedureParameters	Alphanumeric string. 25 characters max.	If using Donation ID as the unique identifier for the procedure setup, populate the DonationSetupId field with the Donation ID. Otherwise, populate the DonationSetupId with the unique identifier for the procedure setup (e.g., 9999999YYYYMM).

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Element	Parent Element	Data Type	Description
InstrumentSerialNumber	ProcedureParameters	String. 20 characters max. Must be a serial number of the instrument registered with DXT.	InstrumentSerialNumber is required. for PUSH mode configurations. When InstrumentSerialNumber is supplied to DXT, DXT will validate the procedure parameters information, and if it is valid, DXT will immediately forward procedure setup file to the specified instrument. Instrument must be in a ready state to accept new parameters, otherwise parameters status will be set to an error state and must be resent from the DMS.
SiteIdentifier	ProcedureParameters	String. 20 characters max. Must be a site name defined in DMS configurations for the site.	The DMS Site Id associated with the Site Name configured in DXT. Refer to DXT Data Management System Administrator's Guide Chapter 4 for the information about DMS configurations. SiteIdentifier is used to properly associate procedure parameters with the DXT site location for the displaying and reporting purposes.
DXTProcedureParameters	ProcedureParameters		Procedure information complex element. The actual content depends on the instrument type as described in the following sections.

User should verify that DMS Site ID on DXT matches Site ID programmed on the DMS. Incorrect DMS Site ID will prevent DXT from processing remote procedure parameters.

All procedure parameters delivery failures, except operator reject, procedure parameter XML validation failure, or invalid Site ID, are non–recoverable and such parameters must be resent from DMS with the new unique DonationSetupID.

ProcedureParameters DXT Response

Upon successful processing, DXT will respond to procedure parameters with the following XML message:

```
<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <ProcedureParameterStatus>
        <DonationSetupId>DONATION_SETUP_ID</DonationSetupId>
        <ProcedureParametersStatusCode>0</ProcedureParametersStatusCode>
    </ProcedureParameterStatus>
</DMSMessage>
```

DONATION_SETUP_ID in the response will always match the one provided in the request.

The status code in the initial response will always be 0 – Parameters are received by DXT. To monitor updates of Procedure Parameters Status updates, refer to Chapter 3.

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Section 2.2 Procedure Parameters for AMICUS Instruments

The DMS will use the Procedure Parameters service to send procedure parameters to DXT. This enables the remote procedure setup in push mode if DXT is licensed for this feature.

Refer to Section 2.1 Procedure Parameters for overview of the message. This section defines content and provides examples for messages specific to AMICUS instruments.

DXTProcedureParameters XSD Definition

This complex type defines the contents of the Procedure Parameters structure as shown below.

```
<xsd:complexType name="DXTProcedureParameters">
    <xsd:sequence>

        <xsd:element name="ProcedureTypeCode">
            <xsd:simpleType>
                <xsd:restriction base="xsd:integer">
                    <xsd:minInclusive value="1" />
                    <xsd:maxInclusive value="4" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>
        <xsd:element name="DonationId" minOccurs="0" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="15" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="DonorId" minOccurs="0" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="15" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="DonorGenderCode">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:pattern value="[MF]" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="DonorHeight" type="xsd:float">
```


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```
        </xsd:element>

        <xsd:element name="DonorWeight" type="xsd:float">
        </xsd:element>

        <xsd:element name="DonorHematocrit" type="xsd:float">
        </xsd:element>

        <xsd:element name="PlateletPrecount" minOccurs="0">
            <xsd:simpleType>
                <xsd:restriction base="xsd:integer">
                    <xsd:minInclusive value="0" />
                    <xsd:maxInclusive value="999" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

    </xsd:sequence>
</xsd:complexType>
```

DXTProcedureParameters Data Description

The following table explains the content of Procedure Parameters message data for AMICUS:

Element	Parent Element	Data Type	Description
DXTProcedureParameters	ProcedureParameters		Container. This tag is required.
ProcedureTypeCode	DXTProcedureParameters	Non–negative integer	AMICUS procedure to be configured on the instrument. This field is required. Acceptable values: 1 – Single Needle Procedure 2 – Double Needle Procedure
DonationId	DXTProcedureParameters	Alphanumeric string. 15 characters max.	Platelet center generated identifier for platelet donation. This field is required if the instrument is configured for Donation ID entry.
DonorId	DXTProcedureParameters	Alphanumeric string. 15 characters max.	Platelet center generated unique identifier for the donor. This field is required if the instrument is configured for Donor ID entry.
DonorGenderCode	DXTProcedureParameters	String. 1 character max.	Donor's gender. Acceptable values are: M (male) or F (female)
DonorHeight	DXTProcedureParameters	Float	Donor's height, expressed in centimeters.
DonorWeight	DXTProcedureParameters	Float	Donor's weight, expressed in kilograms.
DonorHematocrit	DXTProcedureParameters	Float	Donor's hematocrit.

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Element	Parent Element	Data Type	Description
PlateletPrecount	DXTProcedureParameters	Integer. 0 – 999 range.	Donor's pre–count, expressed in 1000/uL. Pre–count is required for Platelet SN and Platelet DN procedures. Pre–count must be within the limits configured on the instrument for the procedure setup file to be accepted.

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Procedure Parameters Example for AMICUS instruments

Following is an example of complete PUSH mode Procedure Parameters request for the AMICUS instrument with software version 5.13.

```
POST /fresenius-kabi/dxt/dms/procedure-parameters HTTP/1.1
Date: Tue, 30 Oct 2018 16:00:19 GMT
Content-Type: multipart/mixed; boundary=this_is_fresenius_kabi_http_boundary
Cache-Control: no-cache no-transform
FK-SourceId: 1
FK-Protocol-Version: 3.0.0.0
Host: 5.61.64.115
Content-Length: 1037
Expect: 100-continue
Connection: Keep-Alive

--this_is_fresenius_kabi_http_boundary
Content-Type: text/xml; charset=utf-8

<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <ProcedureParameters>
        <DonationSetupId>14EILB093031591</DonationSetupId>
        <InstrumentSerialNumber>AMI1005</InstrumentSerialNumber>
        <SiteIdentifier>Site1</SiteIdentifier>
        <DXTProcedureParameters>
            <ProcedureTypeCode>1</ProcedureTypeCode>
            <DonationId>14EILB093031591</DonationId>
            <DonorId>146551591</DonorId>
            <DonorGendercode>
                F
            </DonorGenderCode>
                <DonorHeight>165.0</DonorHeight>
                <DonorWeight>65.0</DonorWeight>
                <DonorHematocrit>42.0</DonorHematocrit>
                <PlateletPrecount>250</PlateletPrecount>
            </DXTProcedureParameters>
</ProcedureParameters>
</DMSMessage>
```


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```
--this_is_fresenius_kabi_http_boundary--
```

Section 2.3 Instrument Status

The DMS will use the Instrument Status service to determine availability of the instruments for the remote procedure setup. This method prevents collisions when the DXT sends parameters in PUSH mode to the instrument that is currently performing procedure or otherwise unable to accept such parameters.

Instrument Status URL

```
GET /fresenius-kabi/dxt/dms/instrument-status/{SiteId} HTTP/1.1
```

{SiteId} is the DMS Site Id associated with the Site Name configured in DXT. Refer to DXT Data Management System Administrator's Guide Chapter 4 for the information about DMS configurations.

Instrument Status DXT Response

Upon successful processing, DXT will respond to instrument status request with the following XML Message:

```
<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <InstrumentStatus>
        ...
    </InstrumentStatus>
</DMSMessage>
```

InstrumentStatus XSD Definition

This complex type defines the contents and structure of the Instrument Status DXT reply XML message.

```
<xsd:complexType name="InstrumentStatus">
    <xsd:sequence>

        <xsd:element name="SiteIdentifier">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="20" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="Region">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="50" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>
```

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```
        <xsd:element name="SiteName">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="50" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="Instrument" minOccurs="0" maxOccurs="unbounded">
            <xsd:complexType>
                <xsd:sequence>

                    <xsd:element name="InstrumentSerialNumber">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:minLength value="1" />
                                <xsd:maxLength value="20" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>

                    <xsd:element name="InstrumentName">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:minLength value="1" />
                                <xsd:maxLength value="20" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>

                    <xsd:element name="InstrumentTypeCode" type="xsd:unsignedInt" />

                    <xsd:element name="CurrentInstrumentStatusCode"
                            type="xsd:unsignedInt" minOccurs="0" />

                    <xsd:element name="CurrentInstrumentStatusTime"
                            type="xsd:dateTime" minOccurs="0" />

                    <xsd:element name="ElapsedCollectionTime"
                            type="xsd:nonNegativeInteger" minOccurs="0" />

                </xsd:sequence>
            </xsd:complexType>
        </xsd:element>
    </xsd:sequence>
</xsd:complexType>
```


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InstrumentStatus Data Description

The following table explains the content of the Instrument Status DXT reply message as applicable to AMICUS instruments. Refer to the corresponding Data Management supplement for the description of instrument status data for other instruments.

Element	Parent Element	Data Type	Description
InstrumentStatus	DMSMessage		Container. This element is always provided.
SiteIdentifier	InstrumentStatus	String. 20 characters max.	A DMS site name supplied in the request.
Region	InstrumentStatus	String. 50 characters max.	A name of the region configured in DXT.
SiteName	InstrumentStatus	String. 50 characters max.	A name of the site configured in DXT.
Instrument	InstrumentStatus		Container. This element is repeated per instrument configured at the site.
InstrumentSerialNumber	Instrument	String. 20 characters max.	Serial Number of the instrument. This element is always provided.
InstrumentName	Instrument	String. 20 characters max.	Name of the instrument configured in DXT. This element is always provided.
InstrumentTypeCode	Instrument	Integer	Type of the apheresis instrument: 4 – Amicus This element is always provided.
CurrentInstrumentStatusCode	Instrument	Integer	The last known instrument status: 1 – Available. Instrument ready to be setup. 3 – Collecting. Collection has been started on the instrument. 5 – Disconnected. DXT has lost connection with the Y–box. 7 – Prime. Priming has been started on the instrument. 8 – Inactive. Y–box has lost connection with the instrument. 9 – Misconfigured. Serial number configured in DXT and serial number configured in the instrument do not match. 10 – Ready. This element is omitted if there were no communications with the instrument since DXT Instance has started.

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Element	Parent Element	Data Type	Description
CurrentInstrumentStatusTime	Instrument	Integer	Timestamp of the last status update received from the instrument (instrument status 1, 3, 7, 10). The time when DXT lost communication with the instrument (instrument status 5, 8, 9). This element is omitted if there were no communications with the instrument since DXT Instance has started.
ElapsedCollectionTime	Instrument	Integer	Time in seconds since the start of the collection. This element is reported only for instruments with status 3 – Collecting.

If multiple DXT Instances are deployed, the Instrument Status request must be sent to the DXT instance hosting the site in the request. See DXT Data Management System Administrator's Guide Installation section for more information about using multiple DXT Instances.

Instrument Status Response Example

Following is an example of Instrument Status DXT reply:

```
FK-Protocol-Version: 3.0.0.0
FK-DestinationId: 1
Content-Length: 1071
Cache-Control: no-cache, no-transform
Content-Type: text/xml; charset=utf-8
Date: Tue, 26 Mar 2019 16:03:39 GMT

<?xml version="1.0"?>
<DMSMessage>
    <InstrumentStatus>
        <SiteIdentifier>DMSSITE1</SiteIdentifier>
        <Region>DXTREG1</Region>
        <SiteName>SITE1</SiteName>
        <Instrument>
            <InstrumentSerialNumber>AMI5010</InstrumentSerialNumber>
            <InstrumentName>AMI5010</InstrumentName>
            <InstrumentTypeCode>4</InstrumentTypeCode>
            <CurrentInstrumentStatusCode>3</CurrentInstrumentStatusCode>
            <CurrentInstrumentStatusTime>2019-03-26T16:03:14.512Z
           </CurrentInstrumentStatusTime>
            <ElapsedCollectionTime>522</ElapsedCollectionTime>
        </Instrument>
        <Instrument>
            <InstrumentSerialNumber>AMI5007</InstrumentSerialNumber>
            <InstrumentName>AMI5007</InstrumentName>
            <InstrumentTypeCode>4</InstrumentTypeCode>
```


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```
            <CurrentInstrumentStatusCode>1</CurrentInstrumentStatusCode>
            <CurrentInstrumentStatusTime>2019-03-26T16:03:14.65Z
            </CurrentInstrumentStatusTime>
        </Instrument>
    </InstrumentStatus>
</DMSMessage>
```

Section 2.4 DXT Server Response

This section lists possible responses that DXT can return to the requests originated from the DMS.

OK Response

DXT will return the following HTTP Response header when the request was processed successfully:

```
HTTP/1.1 200 OK
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Connection: Close
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

The response may or may not contain XML message, based on the request.

Request Header Validation Error Response

When DXT encounters problems validating the request, the following response will be returned to the sender:

```
HTTP/1.1 400 Bad Request
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>ERROR_CODE</Code>
        <Description>ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

The following table lists Error codes and descriptions for possible errors:

Error Code	Error_Description	Notes

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Error Code	Error_Description	Notes
51000	FK–SourceId Null or Empty	Sender did not provide “FK–SourceId” header element in the request or the content of “FK–SourceId” header element is empty.
51001	FK–ProtocolVersion Null or Empty	Sender did not provide “FK–ProtocolVersion” header element in the request or the content of “FK–ProtocolVersion” header element is empty.
51002	Date Null or Empty	Sender did not provide “Date” header element in the request or the content of “Date” header element is empty.
51003	Content–Type Null or Empty	Sender did not provide “Content–Type” header element in the request or the content of “Content–Type” header element is empty.
51004	Host Null or Empty	Sender did not provide “Host” header element in the request or the content of “Host” header element is empty.
51005	Cache–Control Null or Empty	Sender did not provide “Cache–Control” header element in the request or the content of “Cache–Control” header element is empty.
51006	Bad FK–ProtocolVersion. Expected: expected version	FK–ProtocolVersion provided in the header is not compatible with DXT version installed.
51007	Inconsistent Content–Length Value, actual counted value: actual content length	Content–Length value provided in the header does not match the size of the message received.
51008	Unexpected Content–Type, Expected: Content–Type expected, Received: Content–Type received	Content–Type provided in the header does not match content–type that is expected for the request.
51009	XSD Validation Failure: xml error description	DXT Server has failed to validate the XML message body in the received request. Error description is provided in the “Description” element of the response XML.
51010	Invalid Cache–Control, Expected: Cache–Control expected	Cache–Control provided in the header does not contain “no–cache no–transform”.

Request Time Validation Error Response

To ensure timely processing of all inbound messages, the DXT Server requires the message timestamp be within 2 minutes of the DXT Server time, when the message is received.

```
HTTP/1.1 403 Forbidden
Content–Length: 147
Content–Type: text/xml; charset=utf–8
Date: TIMESTAMP
Cache–Control: no–cache no–transform
FK–SourceId: 0
FK–DestinationId: DATA_DESTINATION
FK–Protocol–Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
```


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```
        <Code>51100</Code>
        <Description>Invalid Date value</Description>
    </ErrorResponse>
</DXTMessage>
```

Request Length Validation Error Response

To ensure the integrity of all inbound messages, the DXT Server requires that each request contains Content–Length header element.

When request's Content–Length header element is missing or empty, the following response will be returned to the sender:

```
HTTP/1.1 411 Length Required
Content–Length: 157
Content–Type: text/xml; charset=utf–8
Date: TIMESTAMP
Cache–Control: no–cache no–transform
FK–SourceId: 0
FK–DestinationId: DATA_DESTINATION
FK–Protocol–Version: 3.0.0.0

 <DXTMessage>
    <ErrorResponse>
        <Code>51200</Code>
        <Description>
            Content–Length Null or Empty
        </Description>
        </ErrorResponse>
 </DXTMessage>
```

Request HTTP Version Validation Error Response

When the DXT Server encounters unsupported HTTP version in the request, the following response will be returned to the sender:

```
HTTP/1.1 505 HTTP Version Not Supported
Date: TIMESTAMP
Cache–Control: no–cache no–transform
FK–SourceId: 0
FK–DestinationId: DATA_DESTINATION
FK–Protocol–Version: 3.0.0.0
```

This response does not contain XML message.

Request Processing Error Response

When DXT encounters problems processing the request, the following response will be returned to the sender:

```
HTTP/1.1 500 Internal Server Error
Content-Length: XML_LENGTH
```

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```
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

```
<DXTMessage>
    <ErrorResponse>
        <Code>ERROR_CODE</Code>
        <Description>ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

The following table lists Error codes and descriptions for possible errors:

Error Code	Error_Description	Notes
50000	Instrument Not Registered	The desired instrument is not registered with DXT or is registered through the DXT Instance other than one the request was sent to.
50001	Database Error	DXT has failed to access the database (read or write the data) to process the request.
50009	Remote Procedure Setup Not Licensed	DXT has received Procedure Parameters request, but Remote Procedure Setup feature is not licensed.
50014	Site ID Not Valid	SiteIdentifier element in Procedure Parameters request does not match Site ID configured in DXT for any site location. Or SiteId field in Instrument Status Request URL is not one of the Site Identifier entries configured in Site DMS configuration.
50015	Duplicate Donation Setup ID	DonationSetupId element in Procedure Parameters request contains duplicate/previously received identifier.

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Chapter 3 DXT Client

This chapter defines messages that DXT sends to external data management systems.

The destination URL is configurable through DXT User Interface for each data management system DXT communicates with. Refer to DXT Data Management System Administrator's Guide Chapter 4 – DXT Configurations on how to configure DMS interfaces.

For possible error responses from DXT refer to Error Handling section at the end of this chapter.

Section 3.1 Electronic Records for AMICUS Instruments

DXT will use the Electronic Record service to provide the procedure record data to the DMS for records received from AMICUS instruments, if DXT is licensed for the electronic record feature and DMS is configured on DXT UI.

Electronic Record URL

```
POST {Electronic Record URL}/procedure-record HTTP/1.1
```

DXT will send procedure records to the address configured as "Electronic Record URL" through DXT User Interface.

ElectronicRecord XSD Definition

This complex type defines the contents of the Electronic Record structure. Due to the complexity, the definition is broken into several sub–types, each one presented separately.

```
<xsd:complexType name="ElectronicRecord">
    <xsd:sequence>
        <xsd:element name="Events"
                     minOccurs="0">
            <xsd:complexType>
                <xsd:sequence>
                    <xsd:element name="Event"
                                 minOccurs="0"
                                 maxOccurs="unbounded">
                        <xsd:complexType>
                            <xsd:all>
                                <xsd:element name="EVENTTYPE"
                                             minOccurs="0">
                                    <xsd:simpleType>
                                        <xsd:restriction base="xsd:string">
                                            <xsd:maxLength value="30" />
                                        </xsd:restriction>
                                    </xsd:simpleType>
                                </xsd:element>
                                <xsd:element name="EVENTDATETIME"
                                             type="xsd:dateTime"
                                             minOccurs="0" />
```


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```
                        <xsd:element name="EVENTOPERATOR"
                                     minOccurs="0">
                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="20" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>
                        <xsd:element name="EVENTDESCRIPTION"
                                     minOccurs="0">
                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="200" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>
                    </xsd:all>
                </xsd:complexType>
            </xsd:element>
        </xsd:sequence>
        <xsd:attribute name="EventCount"
                       type="xsd:integer" />
    </xsd:complexType>
</xsd:element>

<xsd:element name="Procedure">
    <xsd:complexType>
        <xsd:all>
            <xsd:element name="PROCEDURERECORDKEY">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="50" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
            <xsd:element name="DONATIONID">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="25" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
            <xsd:element name="PROCEDURE"
                         minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="100" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
            <xsd:element name="PROCEDURECOMPLETED"
                         minOccurs="0">
```


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```
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="5" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>
                <xsd:element name="RECORDTYPECODE"
                             minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:integer">
                            <xsd:enumeration value="1">
                            </xsd:enumeration>
                            <xsd:enumeration value="3">
                            </xsd:enumeration>
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>
                <xsd:element name="PHLEBOTID"
                             minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="20" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>
                <xsd:element name="PROCEDUREEND"
                             type="xsd:dateTime"
                             minOccurs="0" />
                <xsd:element name="OPERATORID"
                             minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="20" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>
                <xsd:element name="PROCEDURESTART"
                             type="xsd:dateTime" />
                <xsd:element name="DATETIME"
                             type="xsd:dateTime"
                             minOccurs="0" />
            </xsd:all>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="DeviceInfo"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:sequence>
                <xsd:element name="INSTRUMENTSERIAL"
                             minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
```


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```
                        <xsd:maxLength value="20" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
            <xsd:element name="SWVERSION"
                         minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="12" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
            <xsd:element name="STATE"
                         minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="10" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
            <xsd:element name="INSTLOCATION"
                         minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="6" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
        </xsd:sequence>
    </xsd:complexType>
</xsd:element>
<xsd:element name="Donor"
             minOccurs="0">
    <xsd:complexType>
        <xsd:all>
            <xsd:element name="DONORID"
                         minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="25" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
            <xsd:element name="DONORGENDER"
                         minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="1" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>
            <xsd:element name="DONORWEIGHT"
```


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```
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="DONORHEIGHT"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:string">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="DONORTYPE"
                minOccurs="0">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:maxLength value="3" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>
        <xsd:element name="DONORHCT"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="DONORGLOBIN"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:float">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="CUFFPRESSURE"
```


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```
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="DIAPRESSURE"
                minOccurs="0"
                maxOccurs="1">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="SYSPRESSURE"
                minOccurs="0"
                maxOccurs="1">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="DONORTEMP"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:float">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="DONORRATE"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
```


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```
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>
        </xsd:all>
    </xsd:complexType>
</xsd:element>
<xsd:element name="Results-Apheresis"
             minOccurs="0">
    <xsd:complexType>
        <xsd:all>
            <xsd:element name="ACUSEDRUN"
                         minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                           type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>
            <xsd:element name="ESTACVOL"
                         minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                           type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>
            <xsd:element name="ACINBYPRODUCT"
                         minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                           type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>
            <xsd:element name="ACINSTORAGEPLASMA"
                         minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                           type="xsd:string" />
                        </xsd:extension>
```


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```
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="ACUSEDPRIME"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="WBACRATIO"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="ACUSEDTOT"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="CITRATEINFRATE"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:float">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="COLLECTIONELAPSED"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
```


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```
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="COLLECTIONEND"
                 type="xsd:dateTime"
                 minOccurs="0" />
    <xsd:element name="COLLECTIONSTART"
                 type="xsd:dateTime"
                 minOccurs="0" />
    <xsd:element name="PLASMAENDPT"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="PLASMAPRODUCTVOL"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="DONORPOSTGLOBIN"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:float">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="DONORPOSTHCT"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
```


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```
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>
<xsd:element name="PLATELETPRECOUNT"
             minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:integer">
                <xsd:attribute name="units"
                               type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>
<xsd:element name="CYCLEVOLUME"
             minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:integer">
                <xsd:attribute name="units"
                               type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>
<xsd:element name="TOTVOLOUTKG"
             minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:float">
                <xsd:attribute name="units"
                               type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>
<xsd:element name="TOTVOLOUT"
             minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:float">
                <xsd:attribute name="units"
                               type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>
<xsd:element name="ESTCOLLTIME"
             minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
```


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```
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="MAXIMUMINLETRATE"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="MAXIMUMRETURNRATE"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="MEANPLATELETVOL"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:float">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="PASSTORAGE"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="PLASMACOLLECTED"
```


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```
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="TOTALPLASMALOSS"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="NONPASSTORAGE"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="PLATELETPOSTCOUNT"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="ESTPLATELETYIELD"
                minOccurs="0">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:float">
                        <xsd:attribute name="units"
                                       type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
```


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```
    </xsd:complexType>
</xsd:element>
<xsd:element name="PRODUCTTYPE"
             minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="50" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>
<xsd:element name="TOTALRBCLOSS"
             minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:integer">
                <xsd:attribute name="units"
                               type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>
<xsd:element name="RBCENDPT"
             minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:integer">
                <xsd:attribute name="units"
                               type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>
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             minOccurs="0">
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                <xsd:attribute name="units"
                               type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>
<xsd:element name="REINFUSIONEND"
             type="xsd:dateTime"
             minOccurs="0" />
<xsd:element name="REINFUSESTART"
             type="xsd:dateTime"
             minOccurs="0" />
<xsd:element name="SALINEUSED"
             minOccurs="0">
    <xsd:complexType>
```


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```
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="PLASMASTORAGEENDPT"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="PLASMASTORAGEVOL"
                 minOccurs="0">
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                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="PLASMATOTALENDPT"
                 minOccurs="0">
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                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="SUGSTORFLUIDVOL"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:integer">
                    <xsd:attribute name="units"
                                   type="xsd:string" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
```


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```
                <xsd:element name="NUMBERPLATELETBAGS"
                             type="xsd:integer"
                             minOccurs="0" />
                <xsd:element name="WBDRAWN"
                             minOccurs="0">
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                                <xsd:attribute name="units"
                                               type="xsd:string" />
                            </xsd:extension>
                        </xsd:simpleContent>
                    </xsd:complexType>
                </xsd:element>
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                             minOccurs="0">
                    <xsd:complexType>
                        <xsd:simpleContent>
                            <xsd:extension base="xsd:integer">
                                <xsd:attribute name="units"
                                               type="xsd:string" />
                            </xsd:extension>
                        </xsd:simpleContent>
                    </xsd:complexType>
                </xsd:element>
                <xsd:element name="VOLUMEPROC"
                             minOccurs="0">
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                        <xsd:simpleContent>
                            <xsd:extension base="xsd:integer">
                                <xsd:attribute name="units"
                                               type="xsd:string" />
                            </xsd:extension>
                        </xsd:simpleContent>
                    </xsd:complexType>
                </xsd:element>
            </xsd:all>
        </xsd:complexType>
    </xsd:element>
    <xsd:element name="SoftGoods"
                 minOccurs="0">
        <xsd:complexType>
            <xsd:all>
                <xsd:element name="ACEXPDATE"
                             type="xsd:date"
                             minOccurs="0" />
                <xsd:element name="ACLOTNBR"
                             minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="20" />
                        </xsd:restriction>
```


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```
    </xsd:simpleType>
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<xsd:element name="ACCODENBR"
             minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>
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             type="xsd:date"
             minOccurs="0" />
<xsd:element name="KITLOTNBR"
             minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>
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             minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>
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             type="xsd:date"
             minOccurs="0" />
<xsd:element name="PASLOTNBR"
             minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>
<xsd:element name="PASCODENBR"
             minOccurs="0">
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        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>
<xsd:element name="RFEXPDATE"
             type="xsd:date"
             minOccurs="0" />
<xsd:element name="RFLOTNBR"
```


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```
                minOccurs="0">
        <xsd:simpleType>
            <xsd:restriction base="xsd:string">
                <xsd:maxLength value="20" />
            </xsd:restriction>
        </xsd:simpleType>
    </xsd:element>
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            <xsd:restriction base="xsd:string">
                <xsd:maxLength value="20" />
            </xsd:restriction>
        </xsd:simpleType>
    </xsd:element>
    <xsd:element name="SALINEEXPDATE"
                type="xsd:date"
                minOccurs="0" />
    <xsd:element name="SALINELOTNBR"
                minOccurs="0">
        <xsd:simpleType>
            <xsd:restriction base="xsd:string">
                <xsd:maxLength value="20" />
            </xsd:restriction>
        </xsd:simpleType>
    </xsd:element>
    <xsd:element name="SALINECODENBR"
                minOccurs="0">
        <xsd:simpleType>
            <xsd:restriction base="xsd:string">
                <xsd:maxLength value="20" />
            </xsd:restriction>
        </xsd:simpleType>
    </xsd:element>
    <xsd:element name="OTHEREXPDATE"
                type="xsd:date"
                minOccurs="0" />
    <xsd:element name="OTHERLOTNBR"
                minOccurs="0">
        <xsd:simpleType>
            <xsd:restriction base="xsd:string">
                <xsd:maxLength value="20" />
            </xsd:restriction>
        </xsd:simpleType>
    </xsd:element>
    <xsd:element name="OTHERCODENBR"
                minOccurs="0">
        <xsd:simpleType>
            <xsd:restriction base="xsd:string">
                <xsd:maxLength value="20" />
            </xsd:restriction>
        </xsd:simpleType>
```


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```
                    </xsd:element>
                </xsd:all>
            </xsd:complexType>
        </xsd:element>

        <xsd:element name="AdditionalData"
                     minOccurs="0"
                     maxOccurs="unbounded">
            <xsd:complexType>
                <xsd:sequence>
                    <xsd:element name="Name">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:maxLength value="50" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>
                    <xsd:element name="Value">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:maxLength value="100" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>
                    <xsd:element name="Units"
                                 minOccurs="0">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:maxLength value="20" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>
                </xsd:sequence>
            </xsd:complexType>
        </xsd:element>
    </xsd:sequence>
    <xsd:attribute name="device"
                   fixed="4"
                   type="xsd:integer" />
</xsd:complexType>
```

ElectronicRecord Data Description

The following table explains the content of ElectronicRecord:

Element	Parent Element	Data Type	Description

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Element	Parent Element	Data Type	Description
ProcedureRecord	DMSMessage	ElectronicRecord	Container. This element is always provided. A “device” attribute identifies the device type: 4 – Amicus
Events	ProcedureRecord		Container for the instrument supplied events. A “count” attribute indicates the number of Event child elements in the container. This element is omitted if instrument did not supply any events within the record.
Event	Events		Container, repeats for each instrument supplied event. All events appear in the sequence in the chronological order.
EVENTTYPE	Event	String. 30 characters max.	Text description of an event reported by the instrument. This element is always provided. PROCSTART (Procedure start) PROCEND (Procedure end) OPERATOR (Operator entry) STATUS (General Procedure Status) NOTE (System note) WARNING (System warning) Note, that all event descriptions are in the instrument’s language. The exact descriptions of the events will change if instrument’s language is changed.
EVENTDATETIME	Event	dateTime	Instrument local date and time of an event occurred during the procedure and reported by the instrument.
EVNETOPERATOR	Event	String. 20 characters max.	The ID of the Operator signed into the instrument at the time of the event occurred.
EVENTDESCRIPTION	Event	String. 200 characters max.	Instrument supplied description of the Event.

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Element	Parent Element	Data Type	Description
RECORDTYPECODE	ProcedureData	Unsigned integer	Type identifier of a procedure record type: 1 – Donation (i.e., procedure during which the donor has been connected to the instrument in the production mode) 3 – Non–donation (i.e., procedure in the production mode that was terminated prior to donor connect)
PHLEBOTID	ProcedureData	String. 20 characters max.	The identifier of the phlebotomist that performed the venipuncture.
PROCEDUREEND	ProcedureData	DateTime: yyyymmddhh mmss	Instrument local date and time when the procedure ended.
OPERATORID	ProcedureData	String. 20 characters max.	The identifier of the last operator entered into the instrument at the end of the procedure.
PROCEDURESTART	ProcedureData	DateTime: yyyymmddhh Mmss	Instrument local date and time when the procedure started.
DeviceInfo	ProcedureRecord		
INSTRUMENTSERIAL	DeviceInfo	String. 20 characters max.	Serial Number of the instrument that performed the procedure. This element is always provided.
SWVERSION	DeviceInfo	String. 12 characters max.	Software version identifier of the device controller that was current during the procedure. This element is always provided.
STATE	DeviceInfo	String. 10 characters max.	Instrument's operating state. Possible values are: EMPTY READY PRIME RUN
INSTLOCATION	DeviceInfo	String. 6 characters max.	The numerical designation for the location of the instrument that performed the procedure.
Donor	ProcedureData		
DONORID	Donor	Alphanumeric string. 15 characters max.	Donor identifier as entered by the operator at the instrument or sent by DMS in the procedure setup file.
DONORGENDER	Donor	String. 1 character.	Donor's gender. Acceptable values are: M (male) or F (female)

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Element	Parent Element	Data Type	Description
DONORWEIGHT	Donor	Integer	Donor's weight as entered by the operator at the instrument or sent by DMS in the remote procedure setup. A "units" attribute indicates units of measurement: lb kg
DONORHEIGHT	Donor	String	Donor's height as entered by the operator at the instrument or sent by DMS in the remote procedure setup. A "units" attribute indicates units of measurement: in ft/in cm
DONORTYPE	Donor	String. 3 characters max.	Donor's ABO type as entered by the operator at the instrument.
DONORHCT	Donor	Integer	Donor's hematocrit as entered by the operator at the instrument or sent by DMS in the remote procedure setup. A "units" attribute indicates units of measurement: %
DONORGLOBIN	Donor	Float	Donor's hemoglobin as entered by the operator at the instrument or sent by DMS in the remote procedure setup. A "units" attribute indicates units of measurement: g/dL
CUFFPRESSURE	Donor	Integer	Donor's cuff blood pressure as entered by the operator at the instrument. A "units" attribute indicates units of measurement: mmHg
DIAPRESSURE	Donor	Integer	Donor's diastolic blood pressure as entered by the operator at the instrument. A "units" attribute indicates units of measurement: mmHg

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Element	Parent Element	Data Type	Description
SYSPRESSURE	Donor	Integer	Donor's systolic blood pressure as entered by the operator at the instrument. A "units" attribute indicates units of measurement: mmHg
DONORTEMP	Donor	Float	Donor's temperature as entered by the operator at the instrument. A "units" attribute indicates units of measurement: °C °F
DONORRATE	Donor	Integer	Donor's heart rate as entered by the operator at the instrument. A "units" attribute indicates units of measurement: bpm
Results–Apheresis	ProcedureData		
ACUSEDRUN	Results–Apheresis	Integer	Amount of anticoagulant used during the procedure. A "units" attribute indicates units of measurement: ml
ESTACVOL	Results–Apheresis	Integer	Approximate amount of anticoagulant to be used during the procedure. A "units" attribute indicates units of measurement: ml
ACINBYPRODUCT	Results–Apheresis	Integer	Amount of anticoagulant in the concurrent plasma product. A "units" attribute indicates units of measurement: ml

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Element	Parent Element	Data Type	Description
ACINSTORAGEPLASMA	Results–Apheresis	Integer	Amount of anticoagulant in the storage fluid. A “units” attribute indicates units of measurement: ml
ACUSEDPRIME	Results–Apheresis	Integer	Amount of anticoagulant used during prime. A “units” attribute indicates units of measurement: ml
WBACRATIO	Results–Apheresis	Integer	Ratio of anticoagulant to whole blood used during the procedure. A “units” attribute indicates units of measurement: 1:X 1:X ratio is 1 part of ACD to X parts of whole blood.
ACUSEDTOT	Results–Apheresis	Integer	Total volume of anticoagulant used during the procedure. A “units” attribute indicates units of measurement: ml
CITRATEINFRATE	Results–Apheresis	Float	Citrate infusion rate used during the procedure. A “units” attribute indicates units of measurement: mg/kg/min
COLLECTIONELAPSED	Results–Apheresis	Integer	Procedure duration between collection start time and collection end time. A “units” attribute indicates units of measurement: min
COLLECTIONEND	Results–Apheresis	DateTime: yyyymmddhh mmss	Instrument local date and time when the collection ended.
COLLECTIONSTART	Results–Apheresis	DateTime: yyyymmddhh mmss	Instrument local date and time when the collection started.

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Element	Parent Element	Data Type	Description
PLASMAENDPT	Results–Apheresis	Integer	Targeted concurrent plasma volume to be collected in the platelet product. A “units” attribute indicates units of measurement: ml
PLASMAPRODUCTVOL	Results–Apheresis	Integer	Volume of concurrent plasma product collected during the procedure. A “units” attribute indicates units of measurement: ml
DONORPOSTGLOBIN	Results–Apheresis	Float	Donor’s hemoglobin after the procedure. A “units” attribute indicates units of measurement: g/dL
DONORPOSTHCT	Results–Apheresis	Integer	Donor’s approximate hematocrit after the procedure. A “units” attribute indicates units of measurement: %
PLATELETPRECOUNT	Results–Apheresis	Integer	Concentration of donor platelets prior to the procedure. A “units” attribute indicates units of measurement: 1000/µL
CYCLEVOLUME	Results–Apheresis	Integer	Maximum volume of whole blood to be processed during the inlet cycles of a single needle procedure. A “units” attribute indicates units of measurement: ml
TOTVOLOUTKG	Results–Apheresis	Float	Donor’s maximum extracorporeal volume (ECV) calculated by platelet estimator. A “units” attribute indicates units of measurement: ml/kg

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Element	Parent Element	Data Type	Description
TOTVOLOUT	Results–Apheresis	Float	Donor's maximum extracorporeal volume (ECV) percentage calculated by platelet estimator. A "units" attribute indicates units of measurement: %
ESTCOLLTIME	Results–Apheresis	Integer	Estimated duration of the procedure. A "units" attribute indicates units of measurement: min
MAXIMUMINLETRATE	Results–Apheresis	Integer	Maximum inlet flow rate of the single needle procedure. A "units" attribute indicates units of measurement: ml/min
MEANPLATELETVOL	Results–Apheresis	Float	Mean platelet volume of the procedure. A "units" attribute indicates units of measurement: femtoliters
PASSTORAGE	Results–Apheresis	Integer	Actual platelet additive solution volume in the storage fluid of the platelet product. A "units" attribute indicates units of measurement: ml
PLASMACOLLECTED	Results–Apheresis	Integer	Total amount of donor's plasma collected for storage and as a product during the procedure. A "units" attribute indicates units of measurement: ml
TOTALPLASMALOSS	Results–Apheresis	Integer	Total absolute plasma loss for a procedure. A "units" attribute indicates units of measurement: ml

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Element	Parent Element	Data Type	Description
NONPASSTORAGE	Results–Apheresis	Integer	Actual plasma volume in the storage fluid of the platelet product. A “units” attribute indicates units of measurement: ml
PLATELETPOSTCOUNT	Results–Apheresis	Integer	Approximate concentration of donor’s platelets after the procedure. A “units” attribute indicates units of measurement: 1000/µL
ESTPLATELETYIELD	Results–Apheresis	Float	Estimated platelet yield in the product. A “units” attribute indicates units of measurement: xe11
PRODUCTTYPE	Results–Apheresis	String. 30 characters max.	Type of product produced by the procedure. “Platelets (PAS)” “Platelets”
TOTALRBCLOSS	Results–Apheresis	Integer	Total absolute red blood cells loss for a procedure. A “units” attribute indicates units of measurement: ml
RBCENDPT	Results–Apheresis	Integer	Targeted concurrent RBC volume to be collected during a single needle procedure. A “units” attribute indicates units of measurement: ml
RBCVOL	Results–Apheresis	Integer	Actual concurrent RBC volume collected during a single needle procedure. A “units” attribute indicates units of measurement: ml
REINFUSIONEND	Results–Apheresis	DateTime: yyyymmddhh mmss	Instrument local date and time when the reinfusion part of the procedure ended.

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Element	Parent Element	Data Type	Description
REINFUSESTART	Results–Apheresis	DateTime: yyyymmddhh mmss	Instrument local date and time when the reinfusion part of the procedure started.
SALINEUSED	Results–Apheresis	Integer	Actual volume of saline used during the procedure. A “units” attribute indicates units of measurement: ml
PLASMASTORAGEENDPT	Results–Apheresis	Integer	Targeted storage fluid volume for the platelet product. This volume includes PAS if PAS is used. A “units” attribute indicates units of measurement: ml
PLASMASTORAGEVOL	Results–Apheresis	Integer	Actual storage fluid volume for the platelet product. This volume includes PAS if PAS is used. A “units” attribute indicates units of measurement: ml
PLASMATOTALENDPT	Results–Apheresis	Integer	Targeted volume of plasma to be collected for storage and as a product during the procedure. A “units” attribute indicates units of measurement: ml
SUGSTORFLUIDVOL	Results–Apheresis	Integer	Suggested approximate amount of storage fluid for the procedure. A “units” attribute indicates units of measurement: ml
WBDRAWN	Results–Apheresis	Integer	Actual volume of whole blood drawn during the procedure. A “units” attribute indicates units of measurement: ml
VOLUMEENDPT	Results–Apheresis	Integer	Estimated whole blood to be processed during the procedure. A “units” attribute indicates units of measurement: ml

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Element	Parent Element	Data Type	Description
VOLUMEPROC	Results–Apheresis	Integer	Volume of whole blood processed during a procedure. A “units” attribute indicates units of measurement: ml
SoftGoods	ProcedureRecord		Container for reporting anticoagulant solution used during the procedure.
ACEXPDATE	SoftGoods	Date: yyyymmdd	The expiration date of the anticoagulant solution used during the procedure.
ACLOTNBR	SoftGoods	String. 20 characters max.	The lot number of the anticoagulant solution used during the procedure.
ACCODENBR	SoftGoods	String. 20 characters max.	The product code of the anticoagulant solution used during the procedure.
KITEXPDATE	SoftGoods	Date: yyyymmdd	The expiration date of the disposable kit used during the procedure.
KITLOTNBR	SoftGoods	String. 20 characters max.	The lot number of the disposable kit used during the procedure.
KITCODENBR	SoftGoods	String. 20 characters max.	The product code of the disposable kit used during the procedure.
PASEXPDATE	SoftGoods	Date: yyyymmdd	The expiration date of the platelet additive solution used during the procedure.
PASLOTNBR	SoftGoods	String. 20 characters max.	The lot number of the platelet additive solution used during the procedure.
PASCODENBR	SoftGoods	String. 20 characters max.	The product code of the platelet additive solution used during the procedure.
RFEXPDATE	SoftGoods	Date: yyyymmdd	The expiration date of the replacement fluid used during the procedure.
RFLOTNBR	SoftGoods	String. 20 characters max.	The lot number of the replacement fluid used during the procedure.
RFCODENBR	SoftGoods	String. 20 characters max.	The product code of the replacement fluid used during the procedure.
SALINEEXPDATE	SoftGoods	Date: yyyymmdd	The expiration date of the saline solution used during the procedure.
SALINELOTNBR	SoftGoods	String. 20 characters max.	The lot number of the saline solution used during the procedure.
SALINECODENBR	SoftGoods	String. 20 characters max.	The product code of the saline solution used during the procedure.
OTHEREXPDATE	SoftGoods	Date: yyyymmdd	The expiration date of other product used during the procedure.

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Element	Parent Element	Data Type	Description
OTHERLOTNBR	SoftGoods	String. 20 characters max.	The lot number of other product used during the procedure.
OTHERCODENBR	SoftGoods	String. 20 characters max.	The product code of other product used during the procedure.
AdditionalData	ProcedureRecord		Container for the additional data. This element is omitted if instrument did not supply any additional data within the record.
Name	AdditionalData	String. 50 characters max.	Name of the additional data.
Value	AdditionalData	String. 100 characters max.	Value of the additional data.
Units	AdditionalData	String. 20 characters max.	Units of the additional data.

DXT will forward donation records or non–donation records if the record type is selected during DMS configuration. Refer to DXT Data Management System Administrator's Guide Chapter 4 – DXT Configurations on how to configure DMS interfaces.

Electronic Record Example for Donation Record

Following is an example of complete Procedure Record exchange between DXT and the DMS. The instrument type is Amicus with software version 4.52. The RecordType is 1, i.e., Donation.

```
POST /procedure-record HTTP/1.1
Connection: keep-alive,Keep-Alive
Date: Wed, 05 Jun 2019 15:48:08 GMT
FK-SourceId: 0
FK-Protocol-Version: 3.0.0.0
Cache-Control: no-transform, no-cache
Content-Type: text/xml; charset=utf-8
Host: 172.16.10.23:8060
Content-Length: 19196
```

```
<?xml version="1.0"?>
<DMSMessage>
    <ProcedureRecord>
        <ProcedureRecordAmicus device="4">
            <Events EventCount="58">
                <Event>
                    <EVENTTYPE>PROCSTART</EVENTTYPE>
                    <EVENTDATETIME>2019-03-27T09:53:17</EVENTDATETIME>
                    <EVENTDESCRIPTION>
                        Procedure Start, Ver 4.52 Proc 0, Key 2198940, SN 3383, Instr Nbr 3383
                    </EVENTDESCRIPTION>
                </Event>
                <Event>
                    <EVENTTYPE>OPERATOR</EVENTTYPE>
                    <EVENTDATETIME>2019-03-27T09:53:50</EVENTDATETIME>
                    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
                    <EVENTDESCRIPTION>Operator Number: SMT923</EVENTDESCRIPTION>
                </Event>
```


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```
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T09:53:51</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Install Kit</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:02:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Clamp Tests</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:03:06</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Clamp Tests Passed</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:03:06</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Cassette Valve Tests</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:03:12</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Cassette Valve Tests Passed</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:03:12</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Kit Test</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:03:58</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Kit Test Passed</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:03:58</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Prime</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:08:54</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Detector Test</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Storage Fluid (mL) 285 -&gt; 554</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Plasma Product (mL) 0 -&gt; 450</EVENTDESCRIPTION>
```


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```
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>WB to Process (mL) 0 -&gt; 5420</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Weight (lbs) 0 -&gt; 157</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Hematocrit (%) 0 -&gt; 47</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Pre-count (1000/..L) 0 -&gt; 246</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Yield (.. e11) 7.000 -&gt; 7.400</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>MPV (fL) 9.200 -&gt; 10.100</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Hemoglobin (g/dL) 0.000 -&gt; 15.666</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Height (in) 0 -&gt; 70</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:26</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Gender ()  -&gt; Male</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:33</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Donation I.D. Number: W39851810237920</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:38</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
```


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```
    <EVENTDESCRIPTION>Kit Code: 4R2337 (+M5264R2337)</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:39</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Kit Lot Number: FA17L06103 (+$$8010319FA17L06103)</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:39</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Kit Expiration Date: 20190331 (+$$8010319FA17L06103)
     </EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:47</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>PAS Lot Number: FM17G19048 (*!*013FM17G19048)</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:15:47</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>PAS Expiration Date: JAN19 (*!*014JAN19)</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:16:07</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Phlebotomist Number: SMT923</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:16:07</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Procedure Setup</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:17:16</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Start Manual Needle Prime</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:39:07</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>WB to Process (mL) 5420 -&gt; 5340</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:39:07</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Pre-count (1000/..L) 246 -&gt; 250</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T10:40:04</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Storage Fluid (mL) 554 -&gt; 410</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>OPERATOR</EVENTTYPE>
```


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```
    <EVENTDATETIME>2019-03-27T10:44:31</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Max Return Rate (mL/min) 150 -&gt; 130</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>WARNING</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:10:18</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Detected</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:10:25</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Purge</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:10:49</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Detector Test</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>WARNING</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:10:54</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Purge Malfunction</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:10:57</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Purge</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:11:22</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Detector Test</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:11:30</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Purge</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:11:55</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Air Detector Test</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:12:03</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Resume Procedure</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:12:03</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Procedure Setup</EVENTDESCRIPTION>
</Event>
<Event>
```


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```
    <EVENTTYPE>WARNING</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:12:19</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Centrifuge Line Blockage</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:13:38</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Resume Procedure</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:14:31</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>PPP Reinfusion</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:15:24</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Saline/RBC Infusion</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:17:18</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Procedure Wrap-Up</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:18:10</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Product Transfer</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:20:50</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Procedure Paused</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:22:17</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Resume Procedure</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:22:22</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Procedure Paused</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:22:25</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Resume Procedure</EVENTDESCRIPTION>
</Event>
<Event>
    <EVENTTYPE>STATUS</EVENTTYPE>
    <EVENTDATETIME>2019-03-27T12:24:57</EVENTDATETIME>
    <EVENTOPERATOR>SMT923</EVENTOPERATOR>
    <EVENTDESCRIPTION>Product Transfer Complete</EVENTDESCRIPTION>
</Event>
```


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```
        <Event>
            <EVENTTYPE>STATUS</EVENTTYPE>
            <EVENTDATETIME>2019-03-27T12:27:58</EVENTDATETIME>
            <EVENTOPERATOR>SMT923</EVENTOPERATOR>
            <EVENTDESCRIPTION>Remove Kit</EVENTDESCRIPTION>
        </Event>
        <Event>
            <EVENTTYPE>PROCEND</EVENTTYPE>
            <EVENTDATETIME>2019-03-27T12:31:04</EVENTDATETIME>
            <EVENTOPERATOR>SMT923</EVENTOPERATOR>
            <EVENTDESCRIPTION>Procedure End</EVENTDESCRIPTION>
        </Event>
    </Events>
    <Procedure>
        <PROCEDUREKEY>00338320180327095317</PROCEDUREKEY>
        <DONATIONID>W39851810237920</DONATIONID>
        <PROCEDURE>Single Needle Platelets</PROCEDURE>
        <PROCEDURECOMPLETED>TRUE</PROCEDURECOMPLETED>
        <RECORDTYPECODE>1</RECORDTYPECODE>
        <PHLEBOTID>SMT923</PHLEBOTID>
        <PROCEDUREEND>2019-03-27T12:31:04</PROCEDUREEND>
        <OPERATORID>SMT923</OPERATORID>
        <PROCEDURESTART>2019-03-27T09:53:17</PROCEDURESTART>
    </Procedure>
    <DeviceInfo>
        <INSTRUMENTSERIAL>AMI3383</INSTRUMENTSERIAL>
        <SWVERSION>4.52</SWVERSION>
        <INSTLOCATION>1</INSTLOCATION>
    </DeviceInfo>
    <Donor>
        <DONORID>000402</DONORID>
        <DONORGENDER>M</DONORGENDER>
        <DONORWEIGHT units="lbs">157</DONORWEIGHT>
        <DONORHEIGHT units="ft/in">5' 10"</DONORHEIGHT>
        <DONORHCT units="%">47</DONORHCT>
        <DONORGLOBIN units="g/dL">15.7</DONORGLOBIN>
        <DIAPRESSURE units="mmHg">0</DIAPRESSURE>
        <SYSPRESSURE units="mmHg">0</SYSPRESSURE>
        <CUFFPRESSURE units="mmHg">50</CUFFPRESSURE>
        <DONORTEMP units="..F">0</DONORTEMP>
        <DONORRATE units="bpm">0</DONORRATE>
    </Donor>
    <Results-Apheresis>
        <ACUSEDRUN units="ml">520</ACUSEDRUN>
        <ESTACVOL units="ml">485</ESTACVOL>
        <ACINBYPRODUCT units="ml">70</ACINBYPRODUCT>
        <ACINSTORAGEPLASMA units="ml">23</ACINSTORAGEPLASMA>
        <ACUSEDPRIME units="ml">35</ACUSEDPRIME>
        <WBACRATIO units="1:X">10</WBACRATIO>
        <ACUSEDTOT units="ml">555</ACUSEDTOT>
        <CITRATEINFRATE units="mg/kg/min">1.25</CITRATEINFRATE>
        <COLLECTIONELAPSED units="min.">109</COLLECTIONELAPSED>
        <COLLECTIONEND>2019-03-27T12:14:31</COLLECTIONEND>
        <COLLECTIONSTART>2019-03-27T10:25:00</COLLECTIONSTART>
        <PLASMAENDPT units="ml">450</PLASMAENDPT>
        <PLASMAPRODUCTVOL units="ml">445</PLASMAPRODUCTVOL>
        <DONORPOSTGLOBIN units="g/dL">14.0</DONORPOSTGLOBIN>
        <DONORPOSTHCT units="%">44</DONORPOSTHCT>
        <PLATELETPRECOUNT units="1000/..L">250</PLATELETPRECOUNT>
        <CYCLEVOLUME units="ml">150</CYCLEVOLUME>
        <TOTVOLOUTKG units="ml/kg">1.8</TOTVOLOUTKG>
        <TOTVOLOUT units="%">2.5</TOTVOLOUT>
        <ESTCOLLTIME units="min.">118</ESTCOLLTIME>
        <MAXIMUMINLETRATE units="ml/min.">110</MAXIMUMINLETRATE>
        <MAXIMUMRETURNRATE units="ml/min.">130</MAXIMUMRETURNRATE>
```


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```
                <MEANPLATELETVOL units="femtoliters">10.1</MEANPLATELETVOL>
                <PASSTORAGE units="ml">268</PASSTORAGE>
                <PLASMACOLLECTED units="ml">545</PLASMACOLLECTED>
                <TOTALPLASMALOSS units="ml">535</TOTALPLASMALOSS>
                <NONPASSTORAGE units="ml">143</NONPASSTORAGE>
                <PLATELETPOSTCOUNT units="1000/..L">140</PLATELETPOSTCOUNT>
                <ESTPLATELETYIELD units="xe11">7.4</ESTPLATELETYIELD>
                <PRODUCTTYPE>Platelets (PAS)</PRODUCTTYPE>
                <TOTALRBCLOSS units="ml">29</TOTALRBCLOSS>
                <RBCENDPT units="ml">0</RBCENDPT>
                <RBCVOL units="ml">0</RBCVOL>
                <REINFUSIONEND>2019-03-27T12:17:18</REINFUSIONEND>
                <REINFUSESTART>2019-03-27T12:14:31</REINFUSESTART>
                <SALINEUSED units="ml">563</SALINEUSED>
                <PLASMASTORAGEENDPT units="ml">410</PLASMASTORAGEENDPT>
                <PLASMASTORAGEVOL units="ml">411</PLASMASTORAGEVOL>
                <PLASMATOTALENDPT units="ml">594</PLASMATOTALENDPT>
                <SUGSTORFLUIDVOL units="ml">554</SUGSTORFLUIDVOL>
                <NUMBERPLATELETBAGS>2</NUMBERPLATELETBAGS>
                <WBDRAWN units="ml">5515</WBDRAWN>
                <VOLUMEENDPT units="ml">5340</VOLUMEENDPT>
                <VOLUMEPROC units="ml">5477</VOLUMEPROC>
            </Results-Apheresis>
            <SoftGoods>
                <KITLOTNBR>FA17L06103</KITLOTNBR>
                <KITCODENBR>4R2337</KITCODENBR>
                <PASLOTNBR>FM17G19048</PASLOTNBR>
            </SoftGoods>
            <AdditionalData />
        </ProcedureRecordAmicus>
    </ProcedureRecord>
</DMSMessage>
```

Section 3.2 Procedure Parameter Status

DXT will use the Procedure Parameter Status service to notify DMS on the status of the procedure parameter, if DXT is licensed for the remote procedure setup.

Procedure Parameter Status URL

```
POST {Procedure Parameters URL}/procedure-parameter-status HTTP/1.1
```

DXT will send procedure parameter statuses to the address configured as “Procedure Parameters URL” through DXT User Interface.

ProcedureParameterStatus XSD Definition

This complex type defines the contents of the Procedure Parameters Status structure as shown below.

```
<xsd:complexType name="ProcedureParameterStatus">
    <xsd:sequence>
        <xsd:element name="DonationSetupId">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
```


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```
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="ProcedureParametersStatusCode" type="xsd:unsignedInt" />

        <xsd:element name="ValidateFailureDetail" type="xsd:string" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>
```

ProcedureParameterStatus Data Description

The following table explains the content of ProcedureParameterStatus:

Element	Parent Element	Data Type	Description
ProcedureParameterStatus	DMSMessage		Container. This tag is required.
DonationSetupId	ProcedureParameter Status	Alphanumeric string. 25 characters max.	Same as the DonationSetupId field, which was sent from DMS in ProcedureParameter message. This element is always provided.
ProcedureParametersStatusCode	ProcedureParameter Status	Unsigned Integer	Parameters status: 1 – parameters were received by the instrument; 2 – parameters were accepted by the instrument; 3 – parameters were rejected by the instrument. More information may be provided in ValidateFailureDetail element; 4 – parameters were accepted and/or confirmed by the operator at the instrument; 5 – parameters were rejected by the operator at the instrument; 8 – instrument is not in a valid state to accept parameters or is not configured to accept parameters in the push mode. If neither confirmation questions nor questionnaire were supplied in procedure setup file, the status will be set to 4 after successful validation by the instrument. This element is always provided.

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Element	Parent Element	Data Type	Description
ValidateFailureDetail	ProcedureParameterStatus	String	Description of the reason for instrument rejection of the parameters. This element is provided only with Status Code 3.

ProcedureParameterStatus Example

Following is an example of complete Procedure Parameter Status between DXT and the DMS

```
POST /procedure-parameter-status HTTP/1.1
Connection: keep-alive,Keep-Alive
Date: Tue, 30 Oct 2018 16:11:22 GMT
FK-SourceId: 0
FK-DestinationId: 1
FK-Protocol-Version: 3.0.0.0
Cache-Control: no-transform, no-cache
Content-Type: text/xml; charset=utf-8
Host: 127.0.0.1:8060
Content-Length: 220

<DMSMessage>
    <ProcedureParameterStatus>
        <DonationSetupId>14EILB093031591</DonationSetupId>
        <ProcedureParametersStatusCode>2</ProcedureParametersStatusCode>
    </ProcedureParameterStatus>
</DMSMessage>
```

Section 3.3 DMS Response

This section lists responses that DXT expects from the DMS.

OK Response

Upon successful processing, the DMS should respond to with the following:

```
Header: HTTP 1.1 200 OK
Content-Length: 0
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0
```

Error Response

When the DMS encounters problems validating the request header, the response to DXT should be formatted as following:

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```
HTTP/1.1 {HTTP_ERROR_CODE} {HTTP_ERROR}
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>DMS_ERROR_CODE</Code>
        <Description>DMS_ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

DXT will treat any non–OK response as an error response. DXT does not specifically act on the type of the error response.

When DXT receives any error response to any message, it will retry sending the message up to three times using the delay programmed for each DMS on DXT User Interface between each retry.

When sending Procedure Record, after all retry attempts are exhausted and transmission is still unsuccessful, DXT will record DMS_ERROR_ CODE and DMS_ERROR_DESCRIPTION of last unsuccessful attempt and will display them along with the record on the undelivered records screen as a failure code. If XML message body is not included in DMS error response or correctly formatted ErrorResponse element is not present in XML message, DXT will record and display HTTP_ERROR_ CODE and HTTP_ERROR instead. Refer to DXT Data Management System User's Guide Chapter 4 – Monitoring for an additional information on handling undelivered records.

When sending Procedure Parameter Status, after all retry attempts are exhausted and transmission is still unsuccessful, DXT will abandon transmission and no further attempts to resend status message will be made.

Over time, this information may be subject to change based upon modification in product manufacturing, blood banking, or critical practice. For additional questions please contact the Clinical Technical Hotline at 1–800–937–5060.

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